



HookBazaar

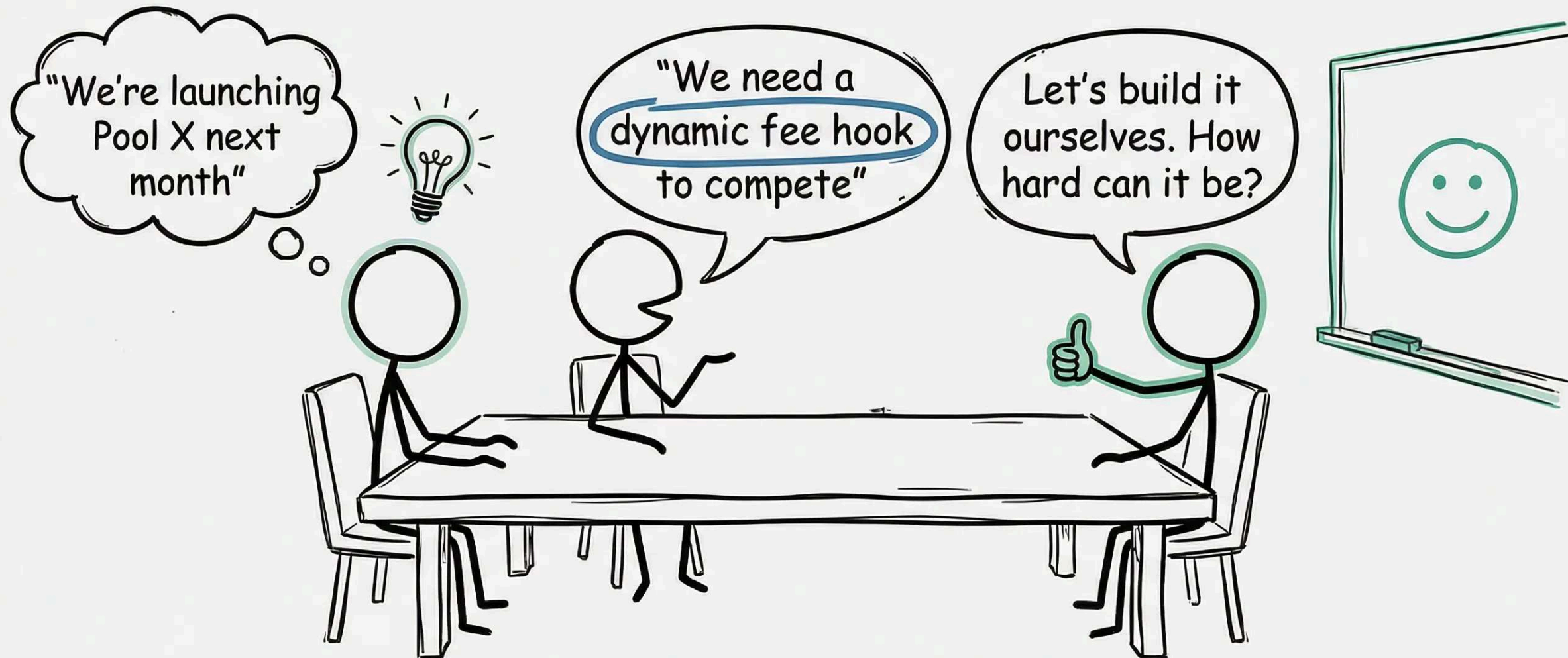
Hook-as-a-Service for Uniswap v4

UHI7 Team: @JMSBPP · @viktor81562 · @iam-Truong

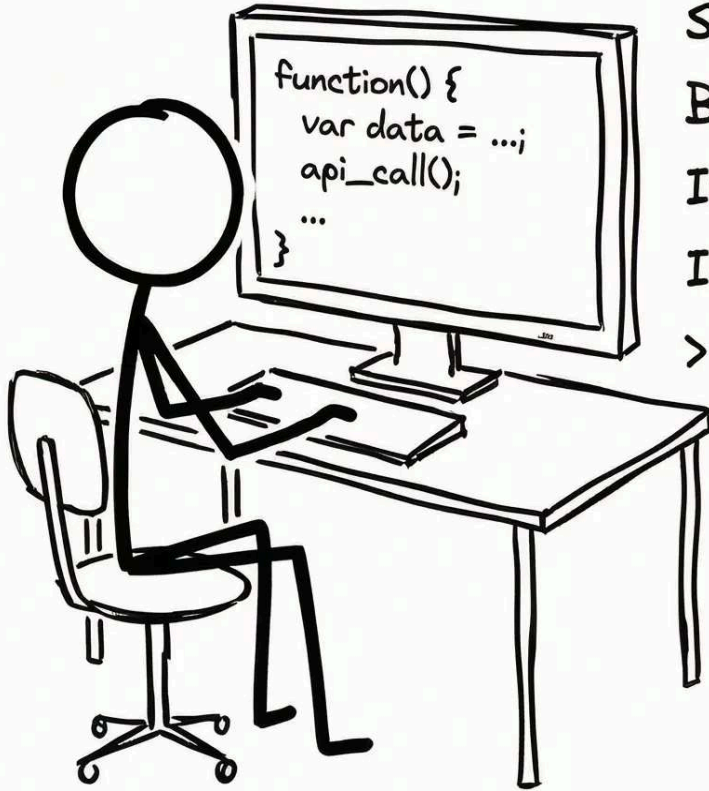
► [Watch Demo on Loom](#)

The Decision

V4



Building Phase - Development Journey



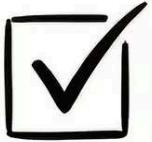
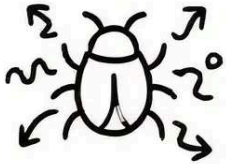
Setting up infra...
Building API...
Integrating AI oracle...
Implementing callbacks...
> deploy hook...

$\$0 \rightarrow \$2,500 \rightarrow \$5,000 \rightarrow \$7,500 \rightarrow \$10,000$

Day 1 $\rightarrow$ Day 7 $\rightarrow$ Day 14 $\rightarrow$ Week 2

$C_{\text{dev}} = \$10,000$ ✓

Testing Phase - Ensuring Quality



Running fuzzers... ↻

Formal verification... ↻

Integration tests... ✓

Edge cases... ✓

✓ 247 tests passed

COST:

\$10,000 → \$13,000

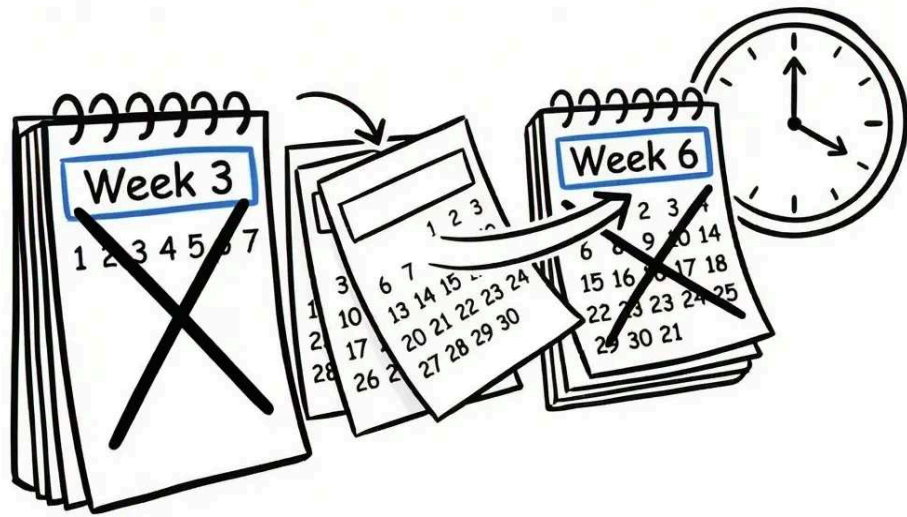
TIME:

Week 2 → Week 3

$C_{\text{test}} = \$3,000$ ✓

Running total: \$13,000

Audit Phase - Security Review



\$13,000

→ \$15,000

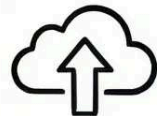
→ \$20,000

→ \$25,000

\$28,000

$C_{\text{audit}} = \$15,000$

Submitting to
auditors...



Waiting...



Review in
progress...



Running total: \$28,000

Total Cost Breakdown

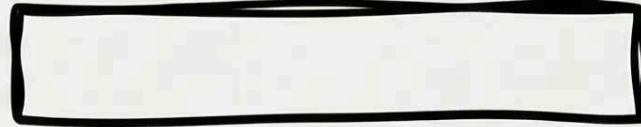
$C_{\text{audit}} = \$15,000$



$C_{\text{test}} = \$3,000$



$C_{\text{dev}} = \$10,000$



$C_{\text{total}} = \$28,000$!

This was a SIMPLE hook. Complex hooks cost \$50k-\$170k+

Register Your Protocol

Connect Wallet



Protocol Name: DeFiSuperStars

Submit Registration



Wallet connected



Protocol registered

Welcome to
Hook Bazaar!

TIME ELAPSED: 30 seconds

[DEMO VIDEO: Registration Flow]

 Search: "dynamic fee"

Dynamic Fee Hook v2.1 ★ 4.8

By: HookDev.eth

Price: \$299

- ✓ Audited by Sherlock
- ✓ 47 pools using this hook

[\[View Details\]](#)

[\[Purchase\]](#)

MEV Shield Hook ★ 4.9

By: SecureHooks.eth

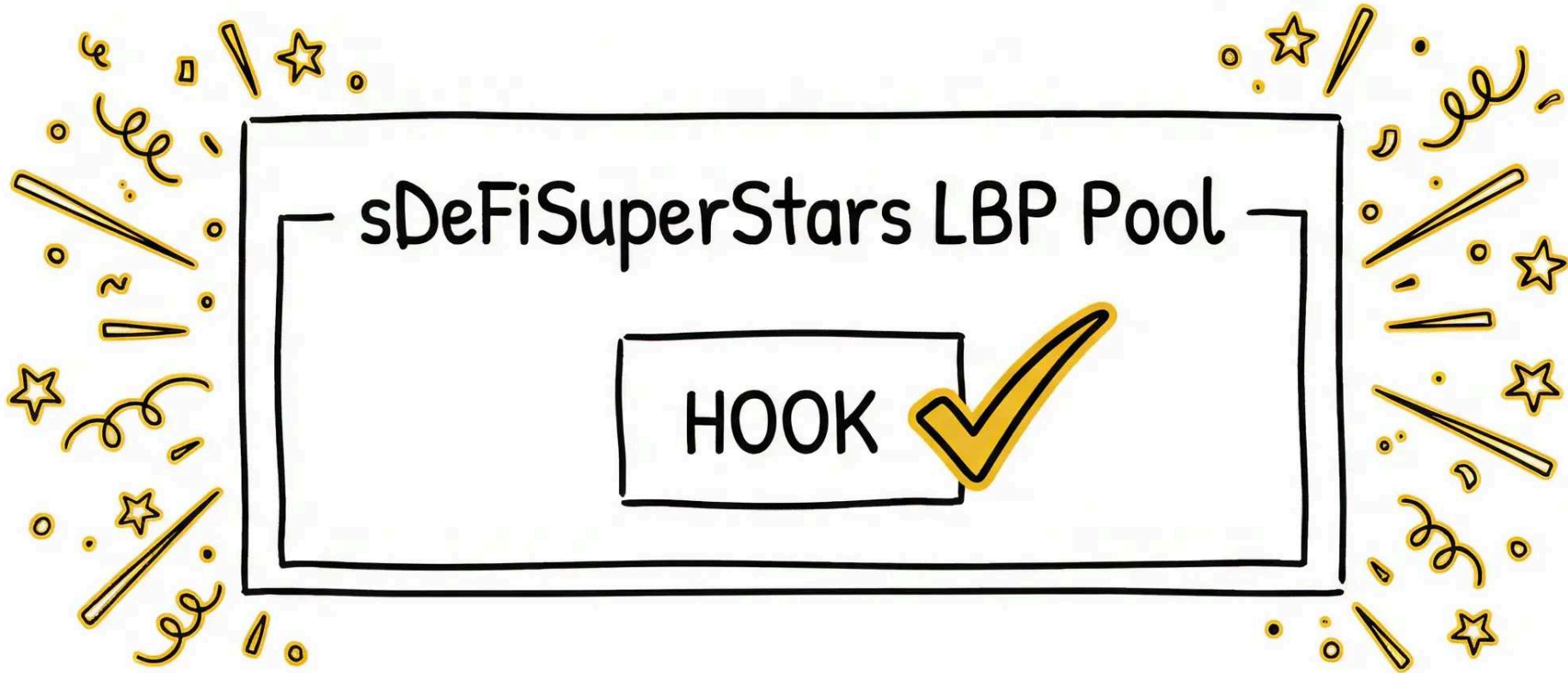
Price: \$499

- ✓ Audited by Trail of Bits

TIME ELAPSED: 2 minutes

[\[DEMO VIDEO: Browse Hooks Flow\]](#)

DEPLOYMENT COMPLETE!



Cost: \$499

Time: 5 minutes

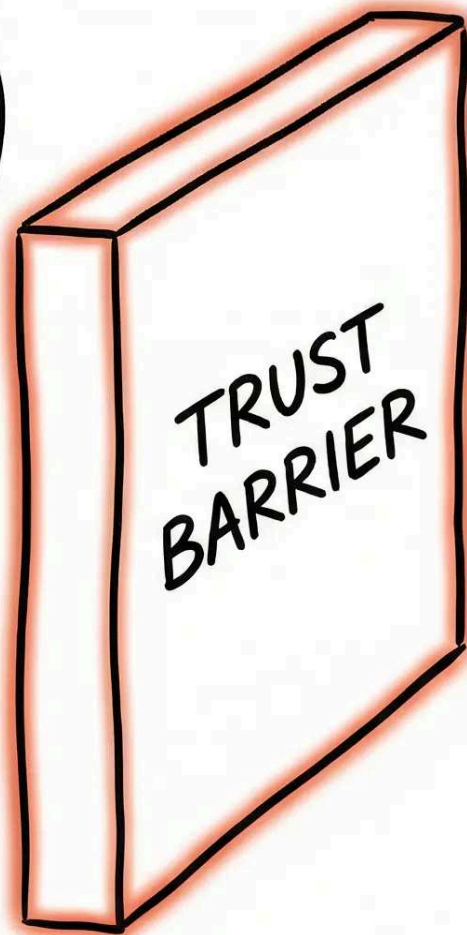
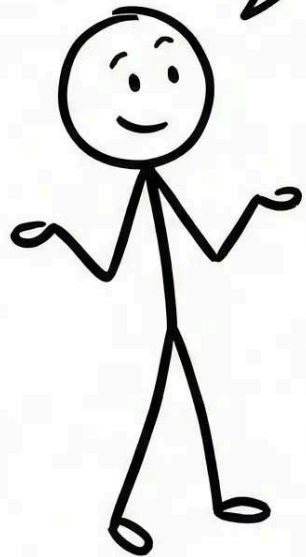
Risk: Minimal
(pre-audited)

Meet @eth1000xSuperDev



A Developer's Distribution Challenge

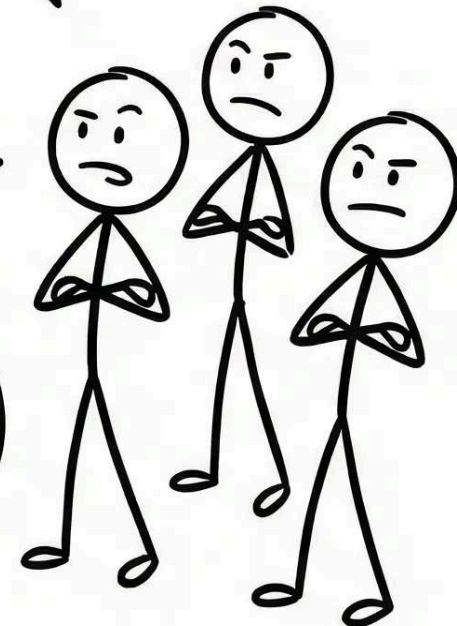
My hook is audited
and battle-tested!



Who audited it?

Never heard
of them.

How do we
know it's secure?



TRUST METER



- Unknown developer: **X**
- Top audit firm: **\$50k+**

~~Free tier
⇒ no income~~

~~Per-transaction
⇒ no enforcement~~

~~Flat fee
⇒ trust needed~~

~~Revenue share
⇒ no infrastructure~~



How do I even get paid?

DEVELOPERS



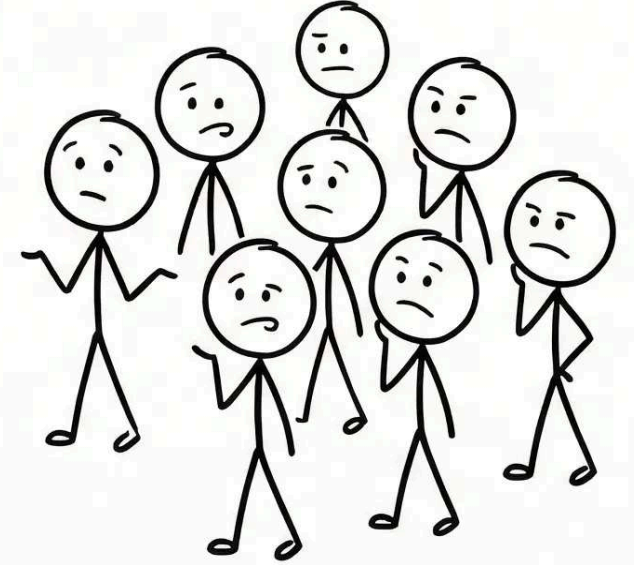
- ✓ Build hooks
- ✗ Can't reach protocols
- ✗ Can't monetize



NO BRIDGE

- No discovery
- No trust system
- No payment rails
- No standards

PROTOCOLS



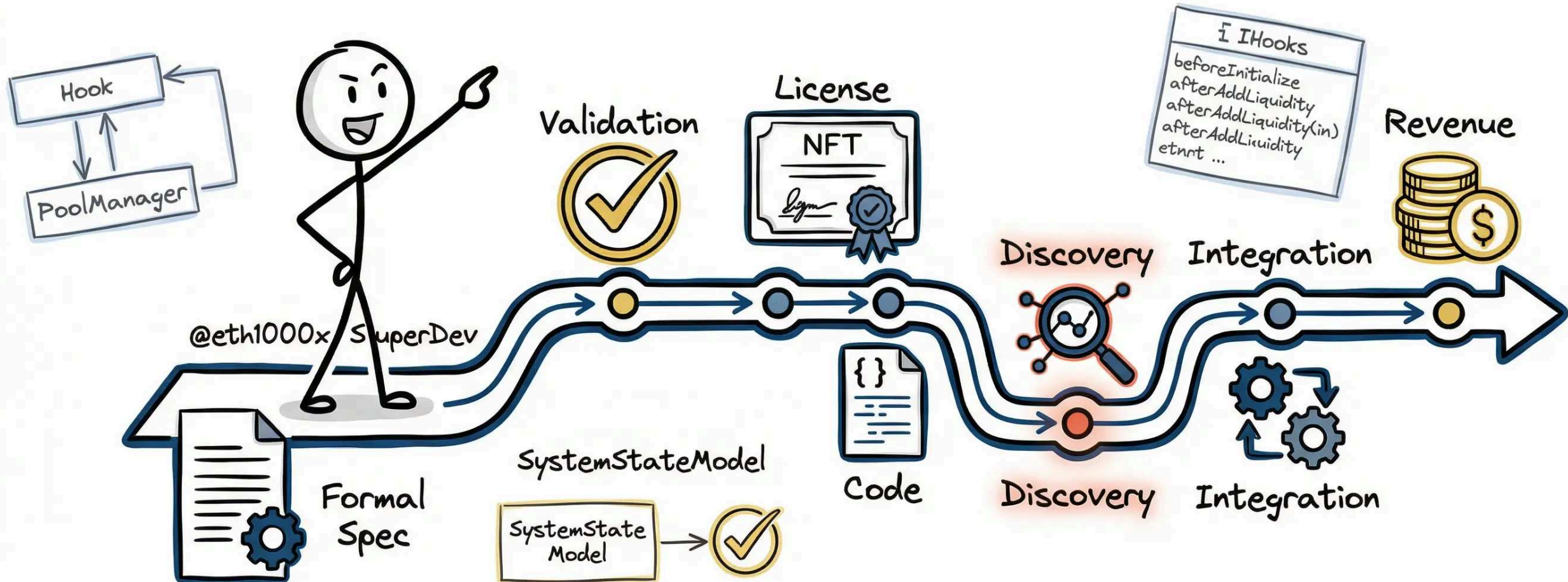
- ✓ Need hooks
- ✗ Can't find developers
- ✗ Can't verify quality

We're both here... but invisible to each other

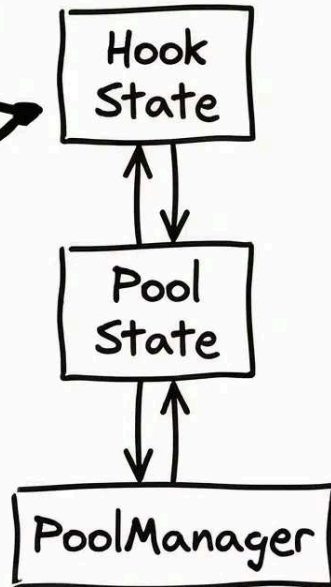


The Hook Developer Journey: From Specification to Discovery

A Technical Guide to Hook Bazaar's Formal Specification System



Building top-down
from IHooks to
achieve my goals



HOOK SPECIFICATION

Target: IHooks entry points

State Variables:

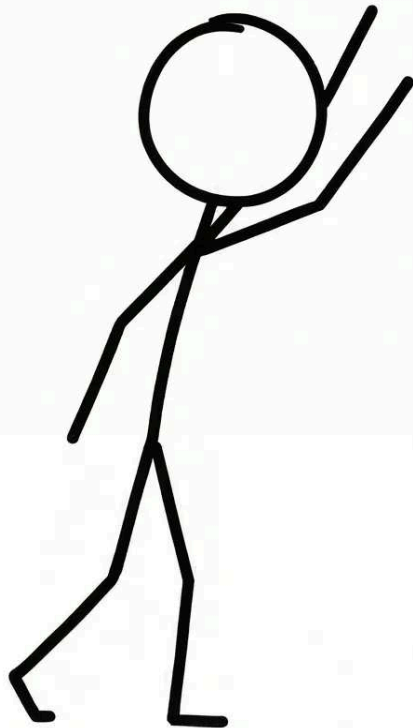
- Maps to `IStateView` canonical vars
- `PoolManager` context interactions

Service Definition:

- `Anti-snipe` protection for LBP pools
- `beforeSwap` callback implementation

Formal Invariants:

- State consistency guarantees
- AMM theory compliance



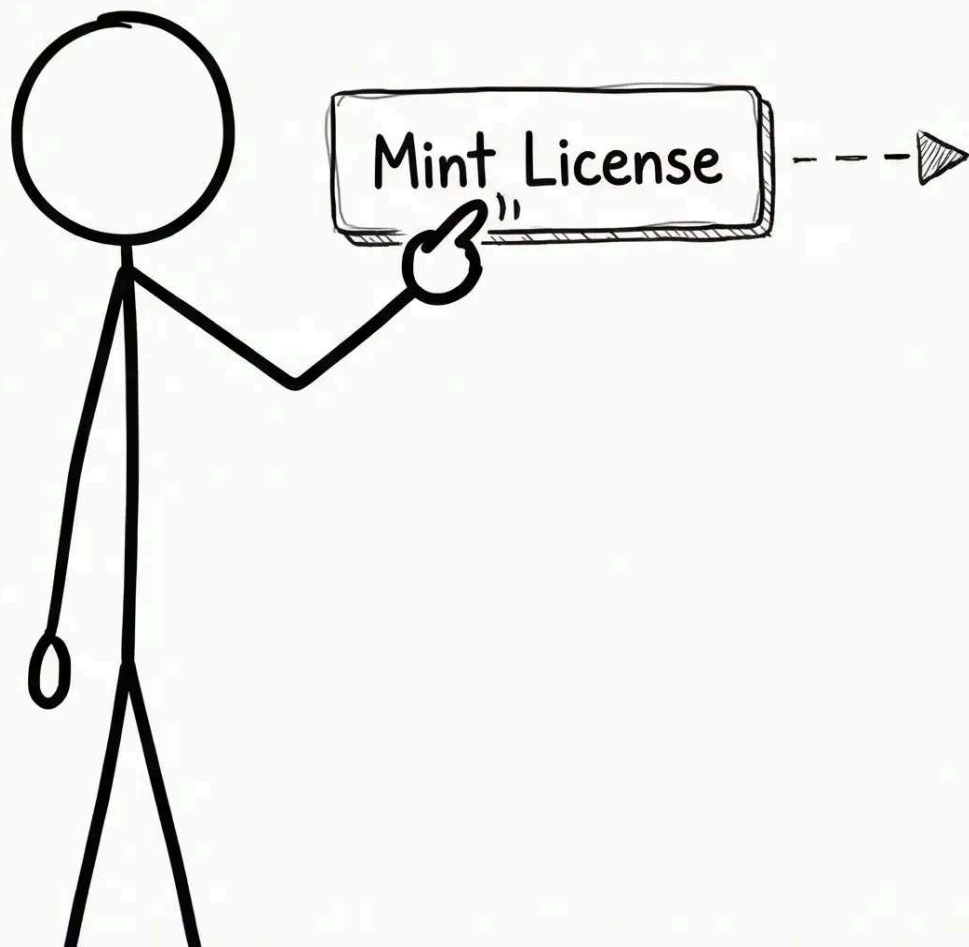
SystemStateModel Compatibility Check

Validating...

- ✓ State variables mapped correctly
- ✓ IHooks interface compliant
- ✓ PoolManager interactions valid
- ✓ AMM theory constraints satisfied
- ✓ No state conflicts detected

RESULT: ✓ COMPATIBLE

Your spec is compatible with
the SystemStateModel!



HOOK MINT LICENSE

License ID: #4721

Developer: @eth1000xSuperDev

Spec Hash: 0x7a3f...

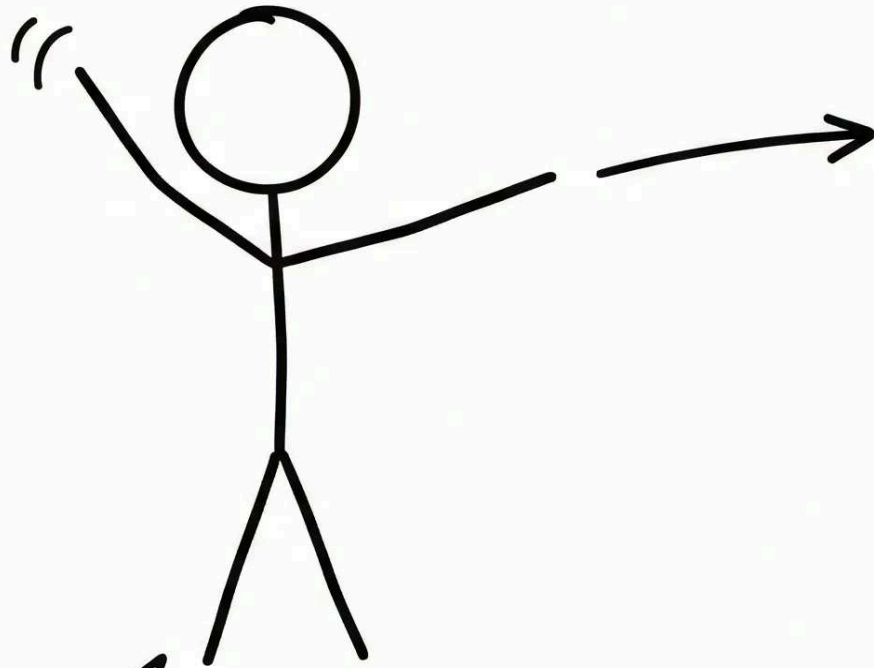
Validated:  SystemStateModel

RIGHTS GRANTED:

-  Submit source code
-  Automatic tagging
-  Metadata generation
-  Marketplace listing

 **Minted!**





The system tagged it automatically and generated the metadata for me!

✓ AntiSnipeHook.sol

SUBMIT WITH HOOKMINTLICENSE:

AUTOMATIC PROCESSING:

- ✓ Code matches spec
- ✓ Compiles successfully
- ✓ Tests pass (247/247)

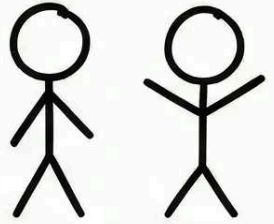
AUTO-GENERATED:

- ✓ Tags: [MEV-Protection] [LBP] [beforeSwap]
- ✓ Metadata: Category, compatibility, docs
- ✓ Integration guide created

STATUS:

✓ **LIVE ON MARKETPLACE**

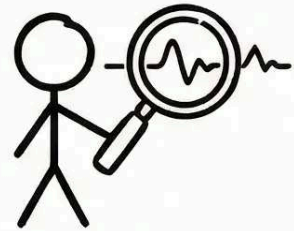
PROTOCOLS



→ DeFiSuperStars wants to integrate

→ AVS Operator verifies requirements ✓

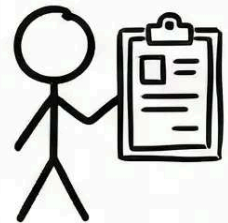
RESEARCHERS



→ AMM Lab validated HookSpec soundness

→ Formal verification adds credibility ✓

AUDITORS



→ Sherlock team reviewing for audit

→ Access to code + spec + validation ✓

DASHBOARD STATS:

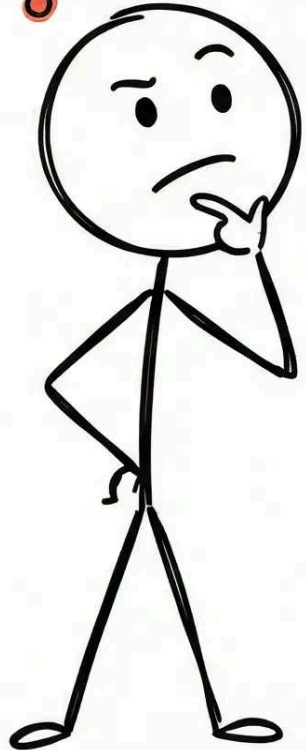
Protocol inquiries: 8

Research citations: 2

Audit requests: 3

Multiple stakeholders, one platform!

But Wait...



Ok, a marketplace for hooks is great, but...

Once the hook is connected to a pool and deployed on-chain...

Can't someone just DECOMPILE the bytecode?

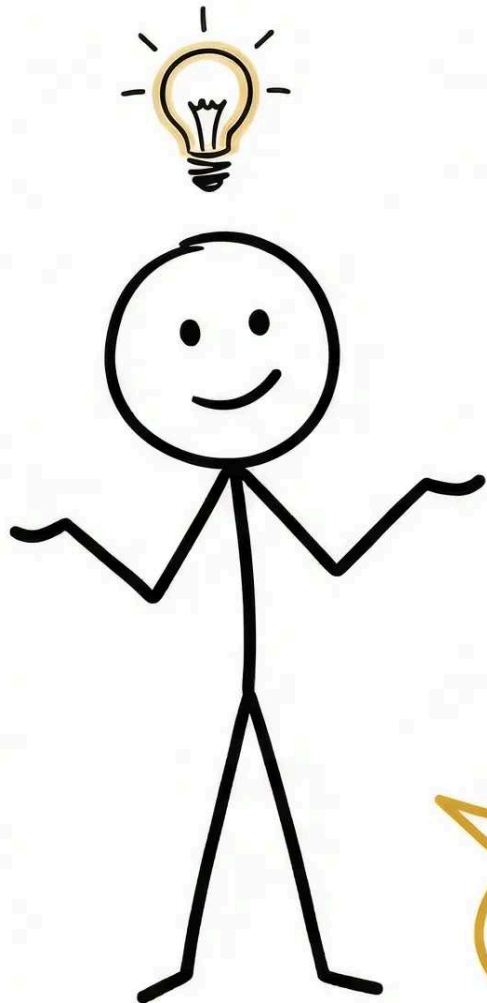
0x608060405234...

Decompiling tools are getting cheaper.

Why pay subscription when you can just... copy?

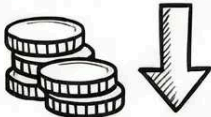
BYTECODE EXPOSED

Fhenix FHE Solution



FULLY HOMOMORPHIC ENCRYPTION

BEFORE:

- Hook Function Calldata: 
0x608060405234...
(readable)
- Decompiling Cost: 
\$100-1,000

AFTER:

- Hook Function Calldata:  [ENCRYPTED]  ?
(unreadable)
- Decompiling Cost: 
\$1,000,000+

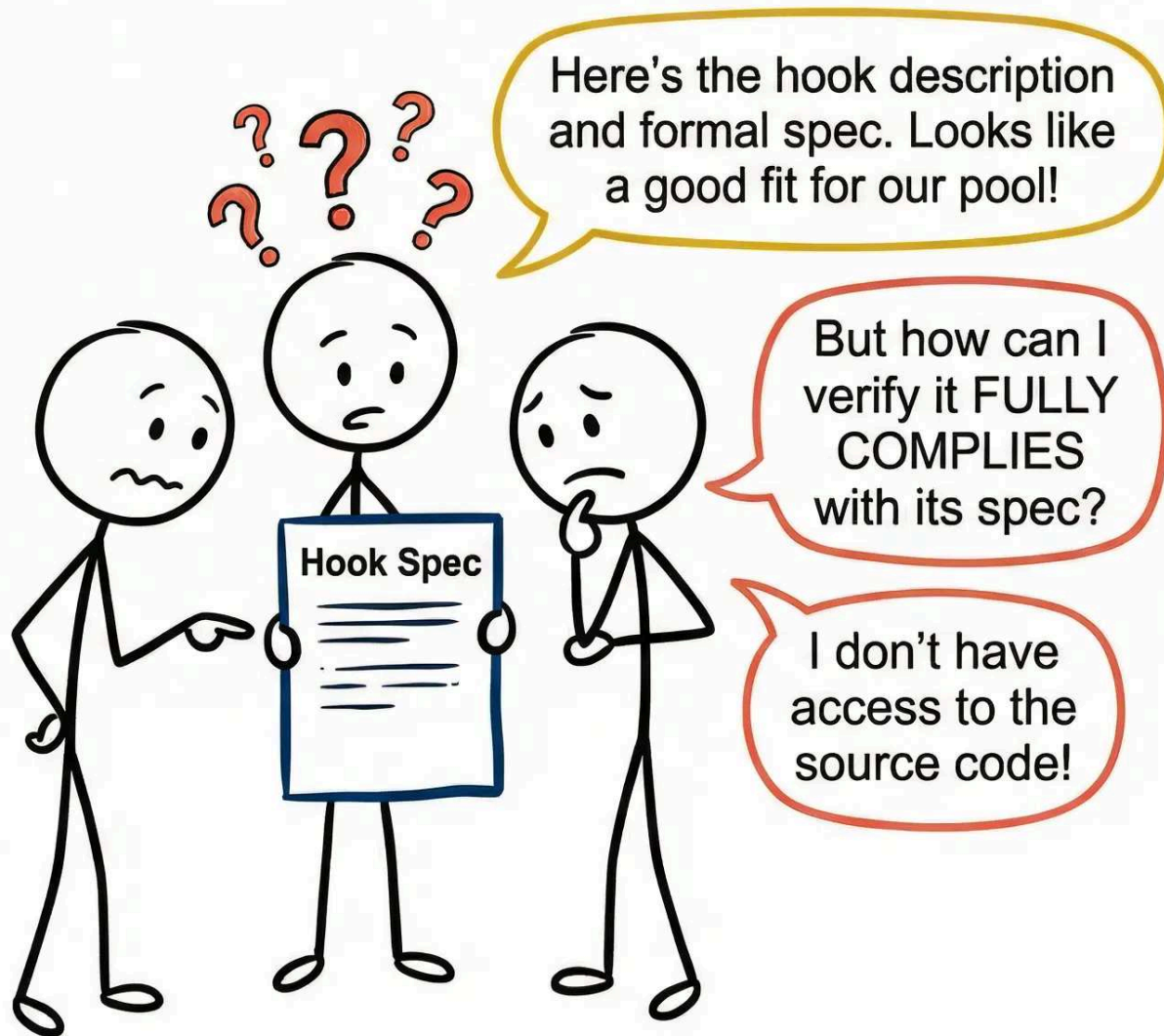
Decompiling cost
goes to the moon!

Fhenix



[DEMO VIDEO: Fhenix
FHE encryption flow] 

Compliance Verification Challenge

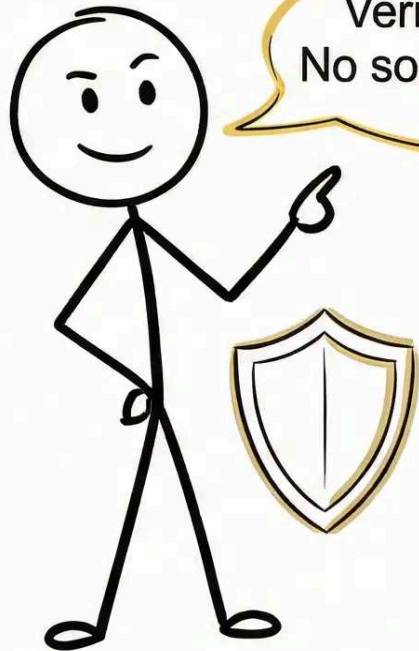


HOOO SPEC DOCUMENT	SOURCE CODE
✓ Can read	✗ No access
✓ Looks good	? Trust issue
✓ Formal description	? Compliance?

How do I know it does what it says?

EigenLayer AVS Attestation

AVS OFF-CHAIN
OPERATOR



Verified compliance.
No source code needed.

SOLUTION

PROVER ATTESTATION:

- ✓ Compliance Score (detailed metrics)
- ✓ Developer Reputation tracking
- ✓ Audit history integration

COMPATIBILITY VERIFICATION:

- ✓ Fuzzing integration tests
- ✓ Protocol-specific semantics check
- ✓ Adapter recommendations

RESULT: Trust without source code access!

[DEMO VIDEO:
AVS Attestation flow]



EigenLayer

- Hook submission, AVS operator verification,
- Compliance score generation, Fuzzing test results, Attestation delivery